AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

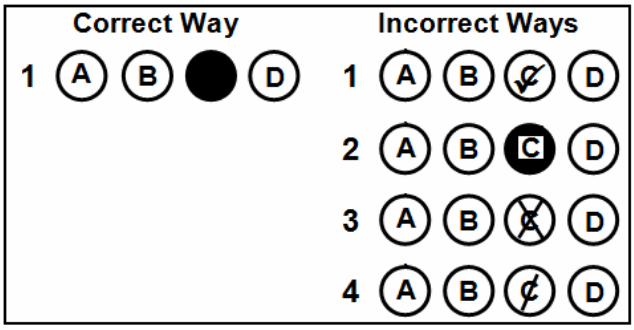
MAY 2013

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. In a simplified form $\frac{3x^2 - 6x}{3x^2}$ is equal to

   A. $1 - 6x$
   B. $1 - 2x$
   C. $\frac{x-2}{x}$
   D. $\frac{x-6}{x}$

2. $P(x, y) = 2x^2 - y$ is a polynomial. Its value at $x = \frac{1}{2}$ and $y = \frac{1}{2}$ is equal to

   A. $-\frac{3}{8}$
   B. $-\frac{1}{4}$
   C. 0
   D. 1

3. If $a - b = 1$ and $a^2 - b^2 = 7$, then $a + b$ is equal to

   A. -7
   B. $-\frac{1}{7}$
   C. $\frac{1}{7}$
   D. 7

4. On factorization of $3x + 6x^2 + 9x^3$, we get

   A. $3x(1 + 2x + 3x^2)$
   B. $3x(1 + 3x + 6x^2)$
   C. $3x(0 + 3x + 6x^2)$
   D. $3x(1 + 2x^2 + 3x^2)$

5. The factors of $1 - x^3$ are

   A. $(1 - x)$ and $(1 + x + x^2)$
   B. $(1 + x)$ and $(1 - x + x^2)$
   C. $(1 + x)$ and $(1 + x + x^2)$
   D. $x, (1 - x)$ and $(1 + x + x^2)$

6. For the zeros of polynomial $x^2 - 4$, which of the following values is true?

I. 2
II. 4
III. -2

A. I only
B. II only
C. I and II only
D. I and III only

7. The highest common factor (H.C.F) of $(x+y)^2$ and $x^2 + xy$ is

A. $(x+y)$
B. $(x+y)^2$
C. $x^2 + xy$
D. $x(x+y)^2$

8. The least common multiple (L.C.M) of 1 and $(x^2 - y^2)$ is

A. 1
B. $(x-y)$
C. $(x+y)$
D. $(x^2 - y^2)$

9. On simplifying $\frac{(a+b)^2}{(a+b)(a-b)(a+b)}$, we get

A. $\frac{1}{a+b}$
B. $\frac{1}{a-b}$
C. $\frac{(a+b)}{a-b}$
D. $\frac{1}{a^2-b^2}$

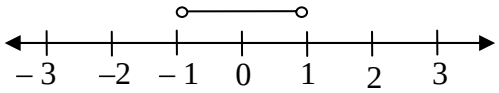
10. The solution set of $x + 2 \leq 2$ is

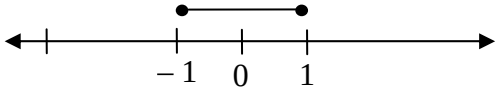
A. $\{\ \}$
B. $\{0\}$
C. $\{0, 1, 2, 3, 4, ...\}$
D. $\{.....-4, -3, -2, -1, 0, ...\}$

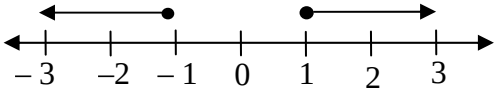
Note: $x \in \omega$

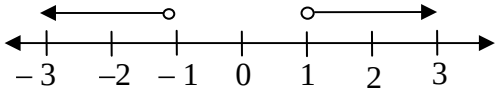
11. The solution set of $|x| < 1$ may be represented on number line as follows

Note: $x \in R$

A. 

B. 

C. 

D. 

12. If the solution set of $\sqrt{x} = -a$ exists, then which of the following statements is correct?

I. $a > 0$
II. $a < 0$
III. $a = 0$

A. I only
B. II only
C. I and III only
D. II and III only

13. The standard form of the quadratic equation $(x-a)(x-b) = 0$ is

A. $x^2 + (a-b)x + ab = 0$
B. $x^2 - (a-b)x - ab = 0$
C. $x^2 - (a+b)x - ab = 0$
D. $x^2 - (a+b)x + ab = 0$

14. For the equation $x^2 - 2x + 2 = 0$, the value of discriminant ($b^2 - 4ac$) is equal to

A. – 12
B. – 4
C. 4
D. 12

15. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is equal to

A. $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

B. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

C. $\begin{bmatrix} -a & 0 \\ 0 & -b \end{bmatrix}$

D. $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

16. $\begin{bmatrix} 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is equal to

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 \end{bmatrix}$

D. $\begin{bmatrix} 2 \end{bmatrix}$

17. Which of the following properties is NOT possessed by matrices of order 2×2 ?

A. $(XY)Z = (YX)Z$

B. $X(YZ) = (XY)Z$

C. $X + (Y + Z) = (X + Y) + Z$

D. $X + (Y + Z) = Y + (X + Z)$

18. For equations $\begin{matrix} 2x - y = 1 \\ 3x - 3y = 2 \end{matrix}$, by Cramer's rule x is equal to

A. $\dfrac{\begin{vmatrix} 2 & -1 \\ 3 & -3 \end{vmatrix}}{\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}}$

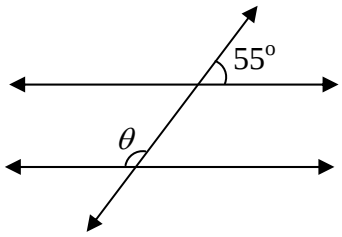
B. $\dfrac{\begin{vmatrix} 1 & -1 \\ 2 & -3 \end{vmatrix}}{\begin{vmatrix} 2 & -1 \\ 3 & -3 \end{vmatrix}}$

C. $\dfrac{\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & -1 \\ 3 & -3 \end{vmatrix}}$

D. $\dfrac{\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}}{\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix}}$

19. In the given figure, θ is equal to

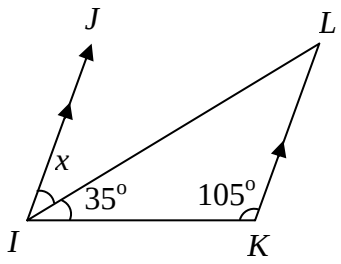
A. 55°
B. 35°
C. 125°
D. 135°



NOT TO SCALE

20. In the given figure, IJ is parallel to KL. The value of x is equal to

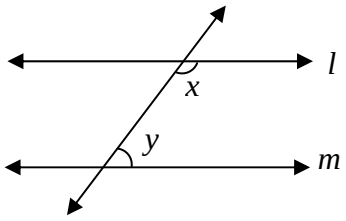
A. 40°
B. 45°
C. 70°
D. 85°



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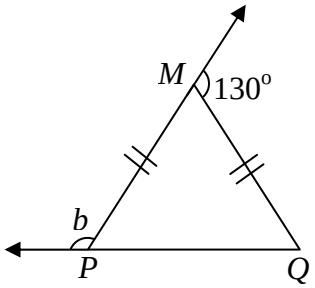
21. In the given figure, l and m are two parallel lines. The value of $x + y$ is equal to

A. 80°
B. 90°
C. 120°
D. 180°



22. In the given triangle MPQ if $MP \cong MQ$, then the value of b will be

A. 110°
B. 115°
C. 120°
D. 130°



NOT TO SCALE

23. Which of the following necessarily passes through the mid-point of the sides of a triangle?

I. Angle bisector
II. Altitude
III. Median

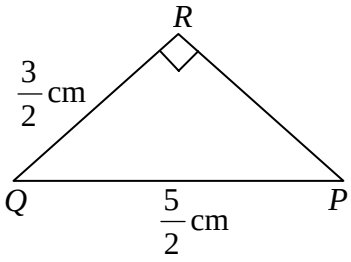
A. I only
B. III only
C. I and II only
D. II and III only

24. In a right angled triangle if the length of the shortest and the longest side are 6 cm and 10 cm respectively, then the length of the remaining side is

A. 6 cm.
B. 7 cm.
C. 8 cm.
D. 9 cm.

25. In the given right angled triangle $m\overline{PR}$ is equal to

A. $\frac{3}{2}$ cm.
B. $\frac{7}{2}$ cm.
C. 2 cm.
D. 4 cm.



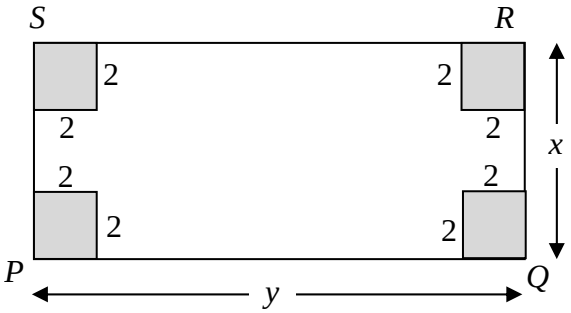
NOT TO SCALE

26. If the radius of a circle is π, then its area will be

A. π^2
B. 2π
C. $2\pi^2$
D. π^3

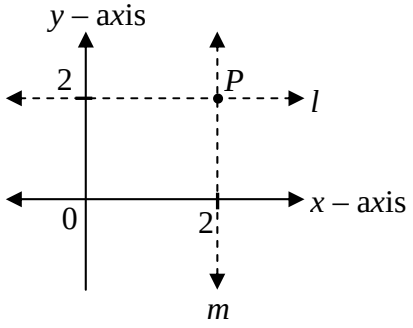
27. In the rectangle *PQRS* the area of unshaded portion is

A. $(x-2)(y-2)$
B. $(x-2)(y-4)$
C. $(x-4)(y-4)$
D. $(x-4)(y-2)$



28. In the given figure, if *P* is the point of intersection of the line *l* and *m*, then $m\overline{OP}$ is equal to

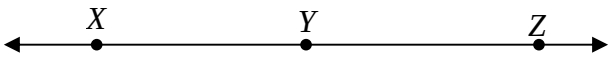
A. 2 units.
B. 4 units.
C. 8 units.
D. $2\sqrt{2}$ units.



NOT TO SCALE

29. If *X*, *Y* and *Z* are collinear points as shown in the figure, then which of the following conditions is correct?

A. $|XY| = |YZ| - |XY|$
B. $|XY| = |XZ| + |YZ|$
C. $|XZ| = |XY| + |YZ|$
D. $|XZ| = |XY| - |YZ|$



30. The distance between the origin *O* (0, 0) and the point (*s*, *t*) is equal to

A. $s + t$
B. $\frac{(s+t)}{2}$
C. $\sqrt{s^2 + t^2}$
D. $\sqrt{s^2 - t^2}$

END OF PAPER